

Decision gate: if PAM does not surpass Tier 2,
Tier 3 targets are adjusted before investing in WP3

Tier 1

Tier 1: Validation

Copy Task ($T=100-5,000$)
Adding Problem ($T=100-5,000$)
MNIST / Omniglot recall
Purpose: Confirm core memory
Baselines: LSTM, GRU, S4

WP1

Tier 2

Tier 2: Field Positioning

ETTh1/h2, Weather, ECL, Traffic
Horizons: {96, 192, 336, 720}
Metrics: MSE, MAE
Purpose: Match SOTA forecasting
Baselines: PatchTST, iTransformer, Mamba

WP1

Tier 3

Tier 3: Core Thesis

sMNIST ($T=784$)
Path-X ($T=16,384$) — gold standard
Memory Maze ($T=2-10K$)
Infinite dSprites ($T=2-50K$)
Purpose: Prove long-range claims
Baselines: S4, Mamba, DNC

WP1-2

Tier 4

Tier 4: Continual Learning

Split MNIST / CIFAR (5 tasks)
Concept drift benchmarks
TSCIL (UCI-HAR, WISDM)
Purpose: Test memory consolidation
Baselines: EWC, SI, replay

WP3

Increasing sequence length, task complexity, and ecological validity →